Exercise 1

Suppose the electron angular distribution in beta decay is

$$W(\theta) = 1 + a\cos\theta\,.$$

Show that under spatial inversion,

$$\mathbf{r} \to -\,\mathbf{r},$$

the angle transforms as

$$\theta \to \pi - \theta,$$

and therefore

$$W(\theta) \to W'(\theta) = 1 - a\cos\theta\,.$$

Explain why a nonzero coefficient a implies parity violation.

Exercise 2

Using

$$k_B = 8.617 \times 10^{-5} \text{ eV/K},$$

calculate $k_B T$ for

$$T = 1 \text{ K}, \qquad T = 10^{-2} \text{ K}, \qquad T = 10^{-3} \text{ K}\,.$$

Comment on why millikelvin temperatures are needed for nuclear polarization.